FML Assignment-1

1. Apply the concept of Machine Learning to identify which type of learning (supervised,unsupervised, or reinforcement) would be suitable for detecting spam emails.

Ans. Based on the concept of Machine Learning presented in the sources, Supervised Learning would be the suitable type of learning for detecting spam emails.

Application of Supervised Learning to Spam Detection

1. Classification Task: Detecting spam emails is primarily a classification problem, which falls under the category of Supervised Learning. Classification involves predicting whether the output variable belongs to a defined category. In the case of spam detection, this involves predicting whether an email is spam (1) or not spam (0).

2. Use of Labeled Data: Supervised learning requires that the machine be trained using data that is well labeled. The training data must contain associated target responses, which can be string labels or numerical values. In the context of spam detection, a dataset would be prepared where emails are explicitly labeled as either "spam" (anomaly) or "nominal" (not spam).

3. Specific Algorithms: Classification algorithms, which are part of supervised learning, are designed for this task. For instance, Logistic Regression is used to predict whether an email is spam (1) or not spam (0) because the target variable is categorical.

4. Real-World Example: The sources note that classifying mail is a common application, exemplified by Gmail, which classifies emails into categories like social, promotions, updates, and forum.

2. Analyze each stage of the Machine Learning life cycle.

Ans. ### 1. Gathering Data

Data Gathering is the initial step in the ML life cycle.

* **Goal:** The primary objective is to identify and obtain all data-related problems.

* **Process:** This step involves identifying various data sources, collecting the necessary data, and integrating the data obtained from these different sources. Data can be collected from various places such as files, databases, the internet, or mobile devices.

* **Outcome:** Successfully performing these tasks yields a set of data, which is also known as a **dataset**. High-quality and accurate data is crucial, as it makes the learning process of the model easier and better. Collecting this data often consumes a huge amount of capital, time, and resources.

2. Data preparation

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After the data has been collected, it must be prepared for subsequent steps

* **Goal:** The purpose of this stage is to put the data into a suitable place and prepare it for use in the machine learning training process. The collected data may be in a raw form and cannot be directly fed to the machine.

* **Sub-processes:** This stage is typically divided into two processes:

1. **Data exploration:** This involves understanding the data's characteristics, format, and quality. A better understanding of the data helps achieve an effective outcome, enabling the identification of correlations, general trends, and outliers.

2. **Data pre-processing:** This prepares the data for its analysis.

3. Data Wrangling (cleaning raw data)

Data wrangling is considered one of the most important steps in the complete ML process.

* **Definition:** This stage is the process of cleaning and converting raw data into a useable format.

* **Addressing Issues:** The collected data is not necessarily always useful and may contain issues that need filtering. Common issues addressed include Missing Values, Duplicate data, Invalid data, and Noise.

* **Action:** Various filtering techniques are used to clean the data during wrangling.

4. Analyse Data

Once the data has been cleaned and prepared, it moves to the analysis stage.

* **Goal:** The aim of this step is to build a machine learning model to analyze the data.

* **Process:** This stage involves the selection of analytical techniques, the building of models, and the review of the results.

* **Techniques:** Machine learning techniques such as **Classification, Regression, Cluster analysis, Association**, and others are selected in this stage.

5. Train the model

This step focuses on optimizing the model's performance.

* **Goal:** To train the model to improve its performance for a better outcome of the problem.

* **Method:** The prepared datasets are used to train the model using various machine learning algorithms.

* **Purpose:** Training is required so that the model can understand the various patterns, rules, and features within the data.

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6. Test the model

Testing occurs after the machine learning model has been trained on a given dataset.

* **Action:** A test dataset is provided to the model to check its accuracy.

* **Outcome:** This step determines the percentage accuracy of the model, assessing whether it meets the requirements of the project or problem.

7. Deployment

Deployment is the final stage of the ML life cycle.

* **Condition:** If the model prepared in the previous stages produces an accurate result, as per the requirements, and operates with an acceptable speed, it is deployed.

* **Action:** The model is deployed into the real-world system.

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3. Analyze the difference between Artificial Intelligence and Machine Learning.

Ans.

AI	ML
AI stands for Artificial intelligence	ML stands for Machine Learning
It is defined acquisition of knowledge intelligence is defined as a ability to acquire and apply knowledge.	It is defined as the acquisition of knowledge or skill
The aim is to increase chance of success and not accuracy.	The aim is to increase accuracy, but it does not care about success.
It work as a computer program that does smart work	It is a simple concept machine takes data and learn from data.
The goal is to simulate natural intelligence to solve complex problem	The goal is to learn from data on certain task to maximize the performance of machine on this task.
AI is decision making.	ML allows system to learn new things from data.
It leads to develop a system to mimic human to respond behave in a circumstances.	It involves in creating self learning algorithms.

AI	ML
AI will go for finding the optimal solution.	ML will go for only solution for that whether it is optimal or not.
AI leads to intelligence or wisdom.	ML leads to knowledge.

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4. Apply your understanding of Machine Learning and Deep Learning. Also, mention the differences between Machine Learning and Deep Learning.

Ans. Machine Learning is defined as the field of study that provides computers with the capability to learn without being explicitly programmed. It gives the computer the ability to learn, making it similar to humans.

Key Principles and Applications of ML:

• Goal: The aim of ML is to learn from data on a certain task to maximize the performance of the machine on that task. ML allows a system to learn new things from data.

• Data Requirement: ML requires datasets so that inferences can be made when predictions can be validated.

• Mathematical Foundation: ML implementations rely on underlying mathematical tools, such as Linear Algebra (concerned with vectors, matrices, and linear transforms).

• Structured Process: ML applications follow a rigorous life cycle involving steps like Gathering Data, Data Wrangling, Training the model, and Testing the model.

• Learning Categories: ML is broadly implemented using three major categories:

◦ Supervised Learning: Algorithms learn from example data that includes associated target responses (labeled data), such as numeric values or string labels, to predict the correct response later. Spam detection is an example of a classification problem, which is a type of supervised learning.

◦ Unsupervised Learning: Algorithms learn from plain examples without any associated response (unlabeled data), determining data patterns or inherent groupings on its own. Clustering algorithms like K-means fall under this category.

◦ Reinforcement Learning: The algorithm is presented with examples that lack labels, but it receives positive or negative feedback (reward or penalty) based on the solution it proposes, similar to learning by trial and error.

Understanding and Application of Deep Learning (DL)

Deep Learning is technically a subset of machine learning. It functions in the same way as ML but possesses different capabilities.

Key Principles and Capabilities of DL:

• Autonomy: In the case of DL, if the model returns an inaccurate prediction, the model does it by himself (corrects the problem), unlike traditional ML where the programmer must fix the problem explicitly.

• Learning Method: A DL model is able to learn from its own method of computing.

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• Handling Unstructured Data: DL architectures, such as the last layers of a convolutional neural network (CNN), can create structure out of unstructured data (like images encoded as pixel sequences), allowing old algorithms that previously only worked on structured data to become useful.

• Interpretation Example: If an ML model is trained to turn on a flashlight only when the word "dark" is used, it will fail if the user says, "I am not able to see anything the light is very dim." A deep learning model would turn the flashlight on because it is able to interpret the underlying meaning beyond the explicit training word.

Characteristic	Machine Learning (ML)	Deep Learning (DL)	Source(s)
Relationship	The larger field of study [1] [15] .	A subset of Machine Learning [15] . [15]	
Model Improvement	Models improve progressively but **still need some guidance** [15] .	Models improve autonomously and are able to learn from their **own method of computing** [16] [17] . [15] [17]	
Handling Inaccuracies	If a model returns an inaccurate prediction, the **programmer needs to fix that problem explicitly** [15] .	If a model returns an inaccurate prediction, the **model does it by himself** [16] . [15] [16]	
Complexity Handling	Can fail in complex tasks where input intent deviates from explicitly trained labels (e.g., the flashlight example) [16] .	Can interpret context and intent beyond explicit labels, leading to successful outcomes in complex scenarios [17] . [16] [17]	

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5. Analyze each stage of the data processing.

Ans. 1. Collection (Gathering Data)

This is the most crucial step when starting with ML.

• Goal and Importance: The primary goal is to identify and obtain all data-related problems. It is vital to have data of good quality and accuracy, as high-quality data makes the learning process of the model easier and better, similar to how students seek the best study material.

• Process: This involves identifying various data sources, collecting the necessary data, and integrating data obtained from different sources. Data can be collected from sources like files, databases, the internet, or mobile devices.

• Outcome and Cost: Performing these tasks yields a dataset. Collecting this data often consumes a huge amount of capital, time, and resources.

2. Preparation

After collection, the data must be prepared for further steps.

• Goal: The purpose is to put the data into a suitable place and prepare it for use in the machine learning training process. This involves collecting datasets from different sources, analyzing them, and constructing a new dataset for further processing and exploration.

• Raw Data: The collected data may be in a raw form which cannot be directly fed to the machine.

• Sub-processes: This stage includes:

◦ Data Exploration: Understanding the characteristics, format, and quality of the data to find correlations, general trends, and outliers.

◦ Data Pre-processing: Preparing the data for subsequent analysis.

• Example: An image might be converted to a matrix of N×N dimensions, where the value of each cell indicates the image pixel.

3. Data Wrangling (Cleaning Raw Data)

Data wrangling is considered one of the most important steps in the complete ML process.

• Goal: It is the process of cleaning and converting raw data into a useable format. This is necessary because the collected data is not necessarily useful and may contain issues.

• Issues Addressed: Wrangling addresses problems such as Missing Values, Duplicate data, Invalid data, and Noise.

• Action: Various filtering techniques are used to clean the data during this stage.

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4. Input

Once prepared, the data may still not be machine-readable, requiring conversion.

• Goal: Convert the prepared data into a machine-readable form.

• Requirements: This conversion task requires high computation and accuracy.

• Examples: Data collected through sources like MNIST Digit data (images), Twitter comments, audio files, or video clips must be converted.

5. Processing (Analysis and Algorithm Execution)

This is the central stage where calculations are performed.

• Goal: This stage requires algorithms and ML techniques to perform the instructions provided over a large volume of data.

• Requirements: Accuracy and optimal computation are needed.

• ML Integration: This corresponds to the Analyze Data step in the ML life cycle, involving the selection of analytical techniques (such as Classification, Regression, Cluster analysis, Association), the building of models, and the review of results.

6. Output

In this stage, the results are delivered from the machine.

• Goal: Results are procured in a meaningful manner that can be easily inferred by the user.

• Form: The final output can be in any desired form, such as reports, graphs, videos, charts, tables, images, and many more, depending on the requirements of the machine and the task being performed.

7. Storage

This is the final step in the data processing sequence.

• Goal: The obtained output, the data model data, and all other useful information are saved.

• Purpose: The saved information is reserved for future use.

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6. Apply your knowledge of matrix operations to calculate the following:

Ans.

Matrix Multiplication: AB

The product AB is calculated by multiplying the rows of A by the columns of B.

$$AB = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} (1 \cdot 9 + 2 \cdot 6 + 3 \cdot 3) & (1 \cdot 8 + 2 \cdot 5 + 3 \cdot 2) & (1 \cdot 7 + 2 \cdot 4 + 3 \cdot 1) \\ (4 \cdot 9 + 5 \cdot 6 + 6 \cdot 3) & (4 \cdot 8 + 5 \cdot 5 + 6 \cdot 2) & (4 \cdot 7 + 5 \cdot 4 + 6 \cdot 1) \\ (7 \cdot 9 + 8 \cdot 6 + 9 \cdot 3) & (7 \cdot 8 + 8 \cdot 5 + 9 \cdot 2) & (7 \cdot 7 + 8 \cdot 4 + 9 \cdot 1) \end{bmatrix}$$

$$AB = \begin{bmatrix} (9 + 12 + 9) & (8 + 10 + 6) & (7 + 8 + 3) \\ (36 + 30 + 18) & (32 + 25 + 12) & (28 + 20 + 6) \\ (63 + 48 + 27) & (56 + 40 + 18) & (49 + 32 + 9) \end{bmatrix}$$

$$\mathbf{AB} = \begin{bmatrix} 30 & 24 & 18 \\ 84 & 69 & 54 \\ 138 & 114 & 90 \end{bmatrix}$$

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Matrix Multiplication: BA

$$BA = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$BA = \begin{bmatrix} (9 \cdot 1 + 8 \cdot 4 + 7 \cdot 7) & (9 \cdot 2 + 8 \cdot 5 + 7 \cdot 8) & (9 \cdot 3 + 8 \cdot 6 + 7 \cdot 9) \\ (6 \cdot 1 + 5 \cdot 4 + 4 \cdot 7) & (6 \cdot 2 + 5 \cdot 5 + 4 \cdot 8) & (6 \cdot 3 + 5 \cdot 6 + 4 \cdot 9) \\ (3 \cdot 1 + 2 \cdot 4 + 1 \cdot 7) & (3 \cdot 2 + 2 \cdot 5 + 1 \cdot 8) & (3 \cdot 3 + 2 \cdot 6 + 1 \cdot 9) \end{bmatrix}$$

$$BA = \begin{bmatrix} (9 + 32 + 49) & (18 + 40 + 56) & (27 + 48 + 63) \\ (6 + 20 + 28) & (12 + 25 + 32) & (18 + 30 + 36) \\ (3 + 8 + 7) & (6 + 10 + 8) & (9 + 12 + 9) \end{bmatrix}$$

$$\mathbf{BA} = \begin{bmatrix} 90 & 114 & 138 \\ 54 & 69 & 84 \\ 18 & 24 & 30 \end{bmatrix}$$

Matrix Addition: $A + B$

Matrix addition is performed by adding corresponding elements.

$$A + B = \begin{bmatrix} 1 + 9 & 2 + 8 & 3 + 7 \\ 4 + 6 & 5 + 5 & 6 + 4 \\ 7 + 3 & 8 + 2 & 9 + 1 \end{bmatrix}$$

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

Matrix Addition: $B + A$

$$B + A = \begin{bmatrix} 9 + 1 & 8 + 2 & 7 + 3 \\ 6 + 4 & 5 + 5 & 4 + 6 \\ 3 + 7 & 2 + 8 & 1 + 9 \end{bmatrix}$$

$$\mathbf{B} + \mathbf{A} = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$